Mathematical Modeling of Autonomous Decision Agents Using Markov Decision Processes with Intent Constraints

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Abstract

This paper presents a rigorous mathematical framework for modeling autonomous decision-making agents operating in stochastic and partially observable environments, with a specific focus on applications in *Agentic Commerce*—the autonomous or semi-autonomous execution of commercial transactions on behalf of human users. Classical Markov Decision Processes (MDPs) provide a foundational structure for sequential decision-making under uncertainty, while their extensions—Partially Observable MDPs (POMDPs) and Constrained MDPs (CMDPs)—address limitations related to hidden state information and operational constraints, respectively. However, these models remain insufficient for capturing explicit human *intent*, which is inherently qualitative, state-dependent, and difficult to encode within traditional reward or cost functions.

To address this gap, we introduce the *Intent-Constrained Markov Decision Process* (ICMDP), a novel extension that incorporates intent as a formal hard constraint on the admissible action space. Unlike reward-based alignment approaches such as Reinforcement Learning from Human Feedback (RLHF), which influence behavior indirectly through reward shaping, the ICMDP framework enforces strict admissibility conditions that guarantee policy compliance with user-defined intent at every decision step. The framework further integrates Large Language Models (LLMs) as probabilistic intent parsers, translating natural language instructions into structured constraint functions that govern agent behavior.

Through theoretical analysis, we show that ICMDP and CMDP are expressively incomparable: ICMDP enforces zero-violation per-step admissibility that no CMDP can replicate, while CMDP captures trajectory-level budget feasibility beyond any state-local filter. A worked example in Agentic Commerce illustrates the framework under partial observability and dynamic intent, and we analyze failure modes, RLHF comparisons, and open computational challenges. The analysis is theoretical; empirical validation remains an open direction.

Keywords: Intent-Constrained MDP (ICMDP), Markov Decision Processes, Constrained MDPs, Partially Observable MDPs, Action Masking, Autonomous Agents, Agentic Commerce, Large Language Models, AI Alignment, Safe Reinforcement Learning.

1 Introduction

Autonomous decision agents are increasingly deployed across robotics, automated finance, logistics optimization, and digital commerce platforms. These agents interact with environments through sequential cycles of observation, action selection, and state transition. Because these interactions unfold over time under uncertainty, rigorous mathematical frameworks are necessary to analyze and optimize such systems.

Markov Decision Processes (MDPs) [10] provide the classical mathematical model for sequential decision-making under uncertainty. In an MDP, an agent observes the current state of the system, selects an action according to a policy, and receives a reward while the environment transitions to a new state. The agent's objective is to find a policy that maximizes cumulative discounted reward over time.

Real-world deployments, however, introduce three challenges that the canonical MDP formalism does not adequately address:

(i) **Partial observability.** Agents rarely have direct access to the true environmental state; instead, they observe noisy projections of it.

(ii) **Operational constraints.** Budget caps, regulatory requirements, and safety limits impose hard feasibility conditions on admissible plans.

(iii) **User intent.** Modern conversational interfaces allow users to specify goals in natural language. These specifications are qualitative and context-sensitive, and cannot be losslessly encoded as scalar reward signals.

The rise of Large Language Models (LLMs) as the interface layer between humans and agentic systems makes the third challenge particularly pressing. An agent executing autonomous purchases on behalf of a user (an instance of *Agentic Commerce*) must interpret intent such as "*buy me the cheapest carbon-neutral option within my* \$100 *budget*" and translate it into a formal decision policy—all while operating under stochastic market conditions.

This paper makes the following contributions:

(1) We provide a self-contained review of MDPs, POMDPs, and CMDPs, locating each framework's expressiveness gap relative to explicit intent modeling (Section 3).

(2) We formally define the *Intent-Constrained Markov Decision Process* (ICMDP), which embeds intent as a state-dependent hard constraint on the admissible action space, and derive its associated Bellman optimality equations (Section 4).

(3) We characterize the LLM as a probabilistic intent parser within this framework, establishing a formal separation of concerns between semantic interpretation and stochastic optimization (Section 5).

(4) We work through a concrete Agentic Commerce scenario, map each architectural layer to a formal component of the ICMDP, and analyze theoretical properties, failure modes, and comparisons to RLHF (Sections 6–7).

2 Related Work

Markov Decision Processes. The theoretical foundations of MDPs and stochastic dynamic programming are due to Puterman [10]. The connection to modern reinforcement learning (RL) algorithms is established in Sutton and Barto [11].

Partial observability. POMDPs were introduced to model environments in which the true system state cannot be directly observed. Kaelbling et al. [7] developed belief-state planning methods that maintain probability distributions over possible states and showed tractable approximation algorithms for finite-horizon problems. Recent work by Lev-Yehudi et al. [9] addresses the computational challenge of deploying POMDP-based systems in real time by proposing probabilistic approximations of observation models with formal degradation guarantees.

Constrained reinforcement learning. CMDPs extend MDPs by constraining expected cumulative costs [3]. Achiam et al. [1] proposed Constrained Policy Optimization (CPO), a practical algorithm for CMDPs that guarantees monotone constraint satisfaction improvement. García and Fernández [5] survey the broader landscape of safe RL techniques.

Alignment and RLHF. The challenge of aligning increasingly capable systems with human preferences is surveyed in Leike et al. [8]. Christiano et al. [4] introduced learning from human feedback as a mechanism to approximate qualitative preferences, forming the conceptual foundation for modern RLHF pipelines.

LLMs for planning. Yao et al. [13] proposed Tree of Thoughts (ToT), enabling LLMs to deliberate over multiple reasoning paths before committing to a decision, demonstrating that sequence models can support structured multi-step planning. Foundational work on the attention mechanism underlying modern LLMs is due to Vaswani et al. [12].

Concurrent and related work on LLM-guided action masking. Zhao et al. [14] propose CAMEL, which uses LLM outputs to dynamically mask the action space during reinforcement learning training via an epsilon-masking schedule that gradually reduces reliance on the LLM guide. Alrashedy et al. [2] introduce Constraints-of-Thought, which restricts the feasible action set during LLM-guided planning by injecting executable constraints that prune search expansions. Both works are closely related to the ICMDP construction.

Positioning of this work. To our knowledge, this work is among the first to formalize intent as a state-dependent hard constraint on the admissible action space within an MDP—which we term an *Intent-Constrained MDP* (ICMDP). The distinguishing features are: a state-dependent filter $\mathcal{I}(s) \subseteq \mathcal{A}$ that enforces per-step admissibility rather than expectation-based penalties; an LLM acting as a probabilistic intent parser that grounds natural-language instructions in a formal constraint structure; and zero violation probability by construction, which expectation-based formulations cannot guarantee. Unlike CAMEL, which uses LLM masking as a training-time exploration heuristic, ICMDP enforces $\mathcal{I}(s)$ as a hard runtime admissibility constraint at every execution step, with formal Bellman analysis and an expressiveness incomparability result (Proposition 4.1). Unlike Constraints-of-Thought, which operates over LLM search expansions, ICMDP is grounded in the standard MDP formalism and admits policy-theoretic guarantees.

3 Mathematical Framework

3.1 Markov Decision Processes

Definition 3.1 (MDP)**.** A *Markov Decision Process* is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma)$, where:

- $\mathcal{S}$ is a finite set of states,
- $\mathcal{A}$ is a finite set of actions,
- $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the transition probability function, with $\mathcal{P}(s' \mid s, a)$ denoting the probability of transitioning to state s' upon taking action a in state s,
- $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function, and
- $\gamma \in [0, 1)$ is the discount factor.

A *stationary policy* $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ specifies the probability $\pi(a \mid s)$ of selecting action a in state s, where $\Delta(\mathcal{A})$ denotes the probability simplex over $\mathcal{A}$. The *state-value function* under policy π is

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \,\middle|\, s_0 = s \right], \tag{1}$$

and the *optimal value function* satisfies the *Bellman optimality equation*:

$$V^*(s) = \max_{a \in \mathcal{A}} \left[R(s, a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a)\, V^*(s') \right]. \tag{2}$$

The corresponding optimal policy is

$$\pi^*(s) = \arg\max_{a \in \mathcal{A}} \left[R(s, a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a)\, V^*(s') \right]. \tag{3}$$

The *action-value function* (Q-function) $Q^\pi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ provides a per-action decomposition of the value function:

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a)\, V^\pi(s'). \tag{4}$$

The optimal Q-function satisfies the *Q-learning Bellman equation*:

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s,a) \max_{a' \in \mathcal{A}} Q^*(s',a'), \tag{5}$$

with $V^*(s) = \max_{a \in \mathcal{A}} Q^*(s,a)$ and $\pi^*(s) = \arg\max_{a \in \mathcal{A}} Q^*(s,a)$. Equation (5) is the basis for model-free Q-learning algorithms [11], which estimate Q^* from sampled transitions without requiring explicit knowledge of $\mathcal{P}$.

While elegant, the MDP formulation assumes full state observability, imposes no operational constraints, and encodes the entire objective within the scalar reward R. These assumptions are violated in Agentic Commerce, where (i) prices and inventory levels are hidden, (ii) budgets and compliance rules must be hard-respected, and (iii) qualitative user preferences cannot be reduced to a single reward signal.

3.2 Partially Observable MDPs

Definition 3.2 (POMDP)**.** A *Partially Observable Markov Decision Process* extends an MDP with an observation model, forming the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{P}, \mathcal{Z}, R, \gamma)$, where:

- $\mathcal{O}$ is a finite set of observations, and
- $\mathcal{Z} : \mathcal{S} \times \mathcal{A} \times \mathcal{O} \to [0,1]$ is the observation probability function, with $\mathcal{Z}(o \mid s', a)$ denoting the probability of observing o after transitioning to s' via action a.

Because the true state is unobservable, the agent maintains a *belief state* $b \in \Delta(\mathcal{S})$, where $b(s)$ is the agent's probability estimate that the current state is s. After taking action a and observing o, the belief is updated via Bayes' rule:

$$b'(s') \propto \mathcal{Z}(o \mid s', a) \sum_{s \in \mathcal{S}} \mathcal{P}(s' \mid s, a)\, b(s). \tag{6}$$

The value function is then defined over the belief simplex:

$$V(b) = \max_{a \in \mathcal{A}} \left[\sum_{s \in \mathcal{S}} b(s)\, R(s,a) + \gamma \sum_{o \in \mathcal{O}} \mathbb{P}(o \mid b, a)\, V(\tau(b,a,o)) \right], \tag{7}$$

where $\tau(b,a,o)$ is the updated belief after action a yields observation o, and $\mathbb{P}(o \mid b, a) = \sum_{s,s'} \mathcal{Z}(o \mid s', a) \mathcal{P}(s' \mid s, a) b(s)$.

POMDP in Agentic Commerce. In an agentic procurement setting, the hidden state $\mathcal{S}$ encodes the true minimum price a vendor will accept, actual inventory levels, and the user's implicit urgency or budget. The agent receives observations $\mathcal{O}$ such as public price listings, historical bid patterns, and stock-out signals. The belief state b is therefore the agent's probabilistic model of the market: for example, "*there is a 60% probability this vendor will drop its price by 5% within two hours, but a 40% risk the item sells out before then.*"

While the POMDP provides a principled model of uncertainty, solving it exactly is PSPACE-hard [7]. The computational burden is exacerbated in real-time commerce, where negotiation must occur within milliseconds. This motivates constraining the policy to a tractable feasible region via CMDP.

3.3 Constrained MDPs

Definition 3.3 (CMDP [3]). A *Constrained Markov Decision Process* augments an MDP with a set of cost functions and associated thresholds: $\mathcal{M}_C = (\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \{C_k\}_{k=1}^n, \{d_k\}_{k=1}^n, \gamma)$, where $C_k : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the k-th cost function and $d_k \in \mathbb{R}$ is its threshold.

The CMDP optimization problem is:

$$\max_\pi \ \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)\right] \quad \text{subject to} \quad \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t C_k(s_t, a_t)\right] \leq d_k, \quad k = 1, \ldots, n. \tag{8}$$

A standard approach linearizes (8) via Lagrange multipliers $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n) \geq \mathbf{0}$ [1, 3]:

$$\mathcal{L}(\pi, \boldsymbol{\lambda}) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)\right] - \sum_{k=1}^{n} \lambda_k \left(\mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t C_k(s_t, a_t)\right] - d_k\right). \tag{9}$$

The optimal policy is obtained by solving the saddle-point problem $\min_{\boldsymbol{\lambda} \geq 0} \max_\pi \mathcal{L}(\pi, \boldsymbol{\lambda})$.

CMDP in Agentic Commerce. Typical constraint dimensions include:

- **Budget**: $\mathbb{E}_\pi[\text{total spend}] \leq \$100$.
- **Vendor trust**: $\mathbb{E}_\pi[\text{fraction of untrusted vendors}] \leq 2\%$.
- **Latency**: expected transaction completion time $\leq 40\,\text{seconds}$.
- **Regulatory compliance**: ISO 20022 header validation must succeed with probability ≥ 0.99.

Combining partial observability with cost constraints yields a framework for *verifiable intent* (VI): the belief state handles probabilistic uncertainty while constraints provide hard operational boundaries.

A critical limitation of CMDP, however, is that constraints are enforced only *in expectation*. The agent may occasionally violate a constraint provided the long-run average remains below the threshold. This is insufficient for qualitative intent specifications such as "*never transact with an unverified seller*", which admit no violations at all.

4 The Intent-Constrained MDP (ICMDP)

4.1 Formal Definition

Definition 4.1 (ICMDP). An *Intent-Constrained Markov Decision Process* is a tuple

$$\mathcal{M}_I = (\mathcal{S},\, \mathcal{A},\, \mathcal{P},\, R,\, \gamma,\, \mathcal{I}),$$

where $(\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma)$ are defined as in a standard MDP and $\mathcal{I} : \mathcal{S} \to 2^{\mathcal{A}}$ is the *intent-constraint function* that maps each state to the set of *intent-admissible* actions.

Remark 4.1 (Notation)**.** The two notations $\mathcal{I} : \mathcal{S} \to 2^{\mathcal{A}}$ and $\mathcal{I}(s) \subseteq \mathcal{A}$ are equivalent and used interchangeably: the former emphasises the functional type, the latter the value at a specific state.

The ICMDP imposes a *hard admissibility condition* on every policy:

$$\pi(a \mid s) = 0 \quad \forall\, a \notin \mathcal{I}(s),\ \forall\, s \in \mathcal{S}. \tag{10}$$

This is strictly stronger than the expectation-based constraints of CMDP (8), which allow occasional violations. The *intent-feasible action set* at state s is

$$\mathcal{A}_{\mathcal{I}}(s) \coloneqq \{a \in \mathcal{A} : a \in \mathcal{I}(s)\}. \tag{11}$$

The *admissible policy class* is

$$\Pi_{\mathcal{I}} \coloneqq \big\{\pi \in \Delta(\mathcal{A})^{\mathcal{S}} : \pi(a \mid s) = 0\ \ \forall\, a \notin \mathcal{I}(s),\ \forall\, s \in \mathcal{S}\big\}. \tag{12}$$

4.2 ICMDP Bellman Equations

The optimal ICMDP value function satisfies the *intent-constrained Bellman optimality equation*:

$$V^*_{\mathcal{I}}(s) = \max_{a \in \mathcal{A}_{\mathcal{I}}(s)} \left[R(s,a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s,a)\, V^*_{\mathcal{I}}(s') \right], \tag{13}$$

with the corresponding optimal intent-constrained policy

$$\pi^*_{\mathcal{I}}(s) = \arg\max\nolimits_{a \in \mathcal{A}_{\mathcal{I}}(s)} \left[R(s,a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s,a)\, V^*_{\mathcal{I}}(s') \right]. \tag{14}$$

The corresponding *intent-constrained action-value function* is

$$Q^*_{\mathcal{I}}(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s,a)\, V^*_{\mathcal{I}}(s'), \quad a \in \mathcal{A}_{\mathcal{I}}(s), \tag{15}$$

so that $V^*_{\mathcal{I}}(s) = \max_{a \in \mathcal{A}_{\mathcal{I}}(s)} Q^*_{\mathcal{I}}(s,a)$ and $\pi^*_{\mathcal{I}}(s) = \arg\max_{a \in \mathcal{A}_{\mathcal{I}}(s)} Q^*_{\mathcal{I}}(s,a)$. The model-free update rule in Section 4.6 estimates $Q^*_{\mathcal{I}}$ directly from sampled transitions.

Remark 4.2 (Reduction to MDP and CMDP)**.** When $\mathcal{I}(s) = \mathcal{A}$ for all s, the ICMDP reduces to a standard MDP. When $\mathcal{I}(s) = \mathcal{A}$ for all s and expectation-based cost constraints are added, the problem reduces to a CMDP, placing MDP $\subset$ CMDP and MDP $\subset$ ICMDP at the base of the hierarchy, with ICMDP and CMDP expressively incomparable above it (Proposition 4.1).

4.3 Feasibility and Infeasibility

A fundamental precondition for meaningful policy optimization under ICMDP is *intent feasibility*: there must exist at least one admissible action in every reachable state.

Definition 4.2 (Intent Feasibility)**.** An ICMDP $\mathcal{M}_I$ is *intent-feasible* if and only if

$$\mathcal{A}_{\mathcal{I}}(s) \neq \emptyset \quad \forall\, s \in \mathcal{S}. \tag{16}$$

If feasibility fails at some state s^*, then $\mathcal{A}_{\mathcal{I}}(s^*) = \emptyset$ and the Bellman recursion (13) is undefined for that state. In this case, the system must either (a) relax the constraint $\mathcal{I}(s^*)$ (*constraint softening*), or (b) surface a *Constraint Violation Warning* to the user, requesting intent re-specification. This introduces a *meta-decision layer* that operates above the ICMDP itself—a novel architectural requirement absent from classical CMDPs.

4.4 Relationship to CMDP: Expressiveness

Proposition 4.1. ICMDP and CMDP are *expressively incomparable*: there exist intent-admissible policy classes $\Pi_{\mathcal{I}}$ that cannot be represented as the feasible set of any CMDP, and there exist CMDP-feasible policy sets that cannot be represented as any ICMDP admissible policy class.

Proof sketch. (*ICMDP $\not\subseteq$ CMDP.*) We assume throughout that CMDP cost functions are bounded: $C_k(s, a) < \infty$ for all s, a, k — the standard assumption in the literature [1, 3]. Consider the intent "*never select action a^- in state s^+*", encoded as $\mathcal{I}(s^+) = \mathcal{A} \setminus \{a^-\}$, so $\Pi_{\mathcal{I}}$ contains only policies with $\pi(a^- \mid s^+) = 0$. Under bounded costs, no CMDP with finite multipliers λ_k can enforce this via (9): the Lagrangian assigns a finite penalty to selecting a^- once, so the CMDP-optimal policy may place nonzero probability on a^- in s^+.

(*CMDP $\not\subseteq$ ICMDP.*) Consider a CMDP constraint $\mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t C(s_t, a_t)] \leq d$ for a cost function $C > 0$ everywhere. The CMDP-feasible set is determined by the entire trajectory distribution: whether action a is permissible in state s depends on the cumulative costs incurred in all other states. No state-dependent filter $\mathcal{I} : \mathcal{S} \to 2^{\mathcal{A}}$ can capture this trajectory-level dependency, because $\mathcal{I}(s)$ is fixed regardless of the history of costs accumulated along the path to s. Hence the CMDP-feasible policy set cannot in general be expressed as any $\Pi_{\mathcal{I}}$.
□ □

4.5 Dynamic Intent

Classical MDP formulations assume that the reward and transition functions are stationary. In long-running Agentic Commerce workflows, user intent may change mid-execution:

$$\mathcal{I}_t(s) \neq \mathcal{I}_{t'}(s) \quad \text{for some } t \neq t'. \tag{17}$$

This introduces a *non-stationary* constrained decision process. The ICMDP framework accommodates two responses:

1. **Constraint injection**: Real-time update of $\mathcal{I}_t(s)$ without full model retraining.

2. **Feasibility alerts**: Proactive notification when market conditions render current intent constraints infeasible (Definition 4.2).

Efficient online algorithms for dynamically reconfigured ICMDPs remain an open research problem. The next section presents solution methods for ICMDP.

4.6 Solutions for ICMDP

The ICMDP formulation modifies the feasible action space rather than the reward or cost structure. As a result, standard dynamic programming and reinforcement learning algorithms can be adapted to operate over the intent-restricted action set $\mathcal{A}_{\mathcal{I}}(s)$. This yields a class of solution methods that we refer to as intent-filtered optimization, in which infeasible actions are pruned prior to policy evaluation and improvement.

4.6.1 Intent-Constrained Value Iteration

The Bellman optimality update for ICMDP (13) can be implemented via a straightforward modification of classical value iteration:

$$V_{k+1}(s) = \max_{a \in \mathcal{A}_{\mathcal{I}}(s)} \left[R(s,a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a)\, V_k(s') \right]. \tag{18}$$

The only modification relative to standard MDP value iteration is the restriction of the maximization operator to the admissible set $\mathcal{A}_{\mathcal{I}}(s)$. This is equivalent to state-dependent action masking applied prior to the Bellman backup [6].

Additionally, the per-iteration computation complexity reduces from $O(|\mathcal{S}||\mathcal{A}|)$ to $O(|\mathcal{S}||\mathcal{A}_{\mathcal{I}}(s)|)$, which can generate efficiency gains when the intent constraints are sufficiently restrictive.

4.6.2 Policy Iteration under Admissibility Constraints

Policy iteration can be adapted by constraining both policy evaluation and improvement steps.

- Policy Evaluation: The evaluation does not change, but it is applied to an admissible policy $\pi \in \Pi_{\mathcal{I}}$.
- Policy Improvement:

$$\pi_{k+1}(s) = \arg\max_{a \in \mathcal{A}_{\mathcal{I}}(s)} Q^{\pi_k}(s,a) \quad \forall\, s \in \mathcal{S}. \tag{19}$$

Unlike CMDPs, no Lagrangian relaxation is required; constraint satisfaction is guaranteed by construction of the policy class $\Pi_{\mathcal{I}}$.

4.6.3 Model-Free RL with Action Masking

In model-free settings, ICMDP can be implemented via action masking in Q-learning and policy gradient algorithms [6]. We note that Huang and Ontañón [6] study masking of *physically impossible* actions (those unavailable in the environment), whereas ICMDP masks *normatively inadmissible* actions—actions that are environmentally possible but excluded by the user's intent constraint. The masking mechanism is identical in both cases; only the source of the mask differs.

Let $\mathcal{M}(s,a)$ denote the action masking function, where $\mathcal{M}(s,a) = 1$ if $a \in \mathcal{A}_{\mathcal{I}}(s)$ and $\mathcal{M}(s,a) = 0$ otherwise.

The Q-learning update estimating $Q^*_{\mathcal{I}}$ (15) is:

$$Q_{\mathcal{I}}(s,a) \leftarrow Q_{\mathcal{I}}(s,a) + \alpha\left[R(s,a) + \gamma \max_{a' \in \mathcal{A}_{\mathcal{I}}(s')} Q_{\mathcal{I}}(s',a') - Q_{\mathcal{I}}(s,a)\right], \quad a \in \mathcal{A}_{\mathcal{I}}(s). \tag{20}$$

For policy gradient methods, the policy is parameterized via masked logits $z_\theta(s,a)$ output by a policy network:

$$\pi_\theta(a \mid s) = \begin{cases} \dfrac{\exp\big(z_\theta(s,a)\big)}{\displaystyle\sum_{a' \in \mathcal{A}_{\mathcal{I}}(s)} \exp\big(z_\theta(s,a')\big)} & \text{if } a \in \mathcal{A}_{\mathcal{I}}(s), \\ 0 & \text{otherwise.} \end{cases} \tag{21}$$

This is equivalent to setting logits of inadmissible actions to $-\infty$ before the softmax, ensuring zero probability mass on inadmissible actions by construction [6].

4.6.4 Feasibility Aware Planning

A critical requirement for all solution methods is intent feasibility (Definition 4.2). In practice, algorithms must incorporate feasibility checks:

- If $\mathcal{A}_{\mathcal{I}}(s) = \emptyset$, the system must trigger constraint relaxation or a fallback policy.
- Maintain a feasibility monitor during planning and execution.

4.6.5 Comparison to CMDP solvers

Unlike CMDP solution methods, which rely on Lagrangian dual optimization and may require iterative tuning of multipliers, ICMDP eliminates constraint violations by restricting the policy space a priori. This removes the need for saddle-point optimization but introduces sensitivity to constraint specification and feasibility.

Section 6 instantiates these solution methods concretely: the Agentic Commerce scenario maps intent-constrained value iteration to the product-catalog state space defined there, the masked Q-learning update (20) to the vendor bidding actions, and the masked softmax policy (21) to the LLM-derived admissibility function (23).

5 LLM as Probabilistic Intent Parser

Let $l \in \mathcal{L}$ denote a natural language instruction provided by the user, where $\mathcal{L}$ is the space of all possible linguistic inputs.

Definition 5.1 (Intent Parser)**.** An *LLM-based intent parser* is a parametric mapping

$$f_\theta : \mathcal{L} \times \mathcal{S} \to 2^{\mathcal{A}}, \qquad (l,s) \mapsto \mathcal{I}(s), \tag{22}$$

where θ denotes the model parameters and the output $\mathcal{I}(s)$ defines the intent-admissible action set for state s given the instruction l.

Unlike traditional parsing systems that rely on predefined grammars, LLMs perform this mapping probabilistically using contextual embeddings and attention mechanisms [12]. The role of f_θ within the ICMDP pipeline is not to optimize actions or policies directly, but to generate a constraint set governing admissible decisions.

This formulation introduces a clean *separation of concerns*:

- f_θ defines the constraint space $\mathcal{I}(s)$ via semantic interpretation of l.
- The ICMDP optimization (13) operates over that constrained space via stochastic dynamic programming.

As a result, improvements in language understanding (better f_θ) do not require changes to the underlying decision algorithms, and vice versa.

Comparison with RLHF. In RLHF [4, 8], human preferences are encoded in a learned reward model $\hat{R}_\phi(s, a)$, and the agent maximizes $\mathbb{E}_\pi[\hat{R}_\phi]$. This is an inherently *soft* mechanism: undesired actions are discouraged but not prohibited. In contrast, the ICMDP intent parser produces *hard* exclusions: $\pi(a \mid s) = 0$ for all $a \notin \mathcal{I}(s)$, regardless of the reward assigned to that action. The two approaches are complementary rather than competing: RLHF can be used to refine behavior *within* the feasible region defined by ICMDP, yielding a hybrid where ICMDP ensures safety and intent compliance and RLHF ensures qualitative optimization within those safe bounds.

Error propagation. The mapping f_θ is probabilistic and subject to misinterpretation. Since constraint errors in ICMDP produce hard exclusions, the downstream effects can be severe: valid actions may be entirely eliminated (*constraint over-tightening*) or prohibited actions may remain admissible (*constraint under-specification*). This highlights the importance of verification mechanisms—e.g., formal consistency checking of the extracted $\mathcal{I}(s)$ against a formal ontology—as a necessary preprocessing step before policy optimization.

6 Worked Example: Agentic Commerce

We illustrate the ICMDP framework with a concrete Agentic Commerce scenario: a user instructs an AI agent, via a conversational interface, to "*Purchase 100 units of Product X from a carbon-neutral supplier, spending no more than \$500, and completing the transaction within 60 seconds.*" The agent must autonomously search product catalogs, evaluate suppliers, negotiate terms, and execute payment, all without further user intervention.

6.1 Architectural Layers

At a structural level, the Agentic Commerce pipeline decomposes into four distinct layers, each corresponding to a formal component of the ICMDP framework (Table 1).

Table 1: Agentic Commerce architectural layers and their formal ICMDP correspondences.

Layer	Component	Formal Role
User / Intent	LLM Intent Parser (f_θ)	Translates natural language l into $\mathcal{I}(s)$; initializes the admissible action space.
Perception	POMDP Belief Updater	Maintains belief $b(s)$ over hidden market states (prices, inventory, seller reliability) via (6).
Decision	ICMDP Solver	Solves (13) over $\mathcal{A}_{\mathcal{I}}(s)$, producing the optimal intent-admissible policy $\pi^*_{\mathcal{I}}$.
Settlement	Payment Switch / Ledger	Executes the chosen action and logs a *certificate of execution*: a mathematical trace proving each action was within $\mathcal{I}(s)$ at every step.

6.2 State and Action Spaces

The state space $\mathcal{S}$ includes tuples of the form $(p_{\text{ask}}, \text{inv}, \text{eco}, \text{rep})$, where p_{ask} is the seller's posted ask price, $\text{inv} \in \{0, 1\}$ indicates availability, $\text{eco} \in \{0, 1\}$ encodes carbon-neutrality certification, and $\text{rep} \in [0, 1]$ is the seller's verifiable reputation score. Although p_{ask} is naturally continuous, we discretize it into a finite set of price bins to satisfy the finite-state assumption of Definition 4.1; this is a standard simplification in applied MDP modeling.

The action space $\mathcal{A}$ includes: $\textsc{Query}(v)$, $\textsc{Bid}(v, p)$, $\textsc{Accept}(v)$, $\textsc{Reject}(v)$, and $\textsc{Pay}(v, \text{rail})$ for vendor v and price p.

6.3 Intent Parsing

The LLM parses the user's instruction into the following formal constraint function:

$$\mathcal{I}(s) = \left\{ a \in \mathcal{A} \;\middle|\; \text{eco}(v(a)) = 1 \;\wedge\; p(a) \leq \$500 \;\wedge\; t_{\text{elapsed}} \leq 60\,\text{s} \right\}, \tag{23}$$

where $v(a)$ is the vendor associated with action a and $p(a)$ is the transaction price. Any action involving a non-certified supplier, an out-of-budget price, or a latency violation is *hard-excluded* from the policy.

6.4 Belief State and POMDP Layer

The true minimum price a vendor will accept is hidden. The agent maintains belief b_t over possible ask prices and updates it upon receiving observations (e.g., a public listing or a counter-offer). For instance, at time t:

$$b_t(\text{ask} = \$4.80/\text{unit}) = 0.60, \quad b_t(\text{ask} = \$5.10/\text{unit}) = 0.40.$$

The agent decides whether to accept immediately or wait for a better quote by evaluating the POMDP value function (7) restricted to $\mathcal{A}_{\mathcal{I}}(s)$.

6.5 Verification Certificate

Because the constraint $\mathcal{I}(s)$ is embedded in the Bellman recursion (13), the agent can produce a certificate of execution for each time step t:

1. **Observed belief**: b_t, representing current market assessment.
2. **Admissible set**: $\mathcal{A}_{\mathcal{I}}(s_t)$, the intent-filtered action space.
3. **Chosen action**: $a_t^* \in \mathcal{A}_{\mathcal{I}}(s_t)$, with proof that $a_t^* \in \mathcal{I}(s_t)$.

This trace provides the user with formal evidence that the agent operated within intent boundaries throughout the transaction—a guarantee unavailable in reward-shaped or expectation-constrained systems.

7 Results and Discussion

7.1 Theoretical Properties of ICMDP

The ICMDP framework introduces two principal structural properties relative to CMDP.

Hard pruning of the policy space. In a standard MDP or CMDP, the feasible policy class is $\Pi = \{\pi : \mathcal{S} \to \Delta(\mathcal{A})\}$ (with expected cost constraints). ICMDP restricts this to $\Pi_{\mathcal{I}}$ as defined in (12). This pruning has two important consequences:

1. **Strong safety and alignment.** Infeasible actions are eliminated prior to optimization rather than penalized post hoc, consistent with the principles of safe RL [5].
2. **Reduced exploration complexity.** The agent need not evaluate actions outside the intent boundary. However, this benefit is conditional on the constraint not being overly restrictive (see feasibility discussion in Section 4.3).

Expressiveness gain over CMDP. CMDP enforces constraints in expectation:

$$\mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t C_k(s_t, a_t) \right] \leq d_k. \tag{24}$$

ICMDP enforces admissibility at every step:

$$\pi(a \,|\, s) = 0 \quad \forall\, a \notin \mathcal{I}(s),\ \forall\, s. \tag{25}$$

Condition (25) captures qualitative intent specifications—such as "*only interact with verified sellers*" or "*avoid environmentally harmful options*"—that cannot be enforced by any CMDP with finite multipliers. Conversely, trajectory-level budget constraints expressible in CMDP have no equivalent state-local representation in ICMDP. Proposition 4.1 formalizes this expressiveness incomparability.

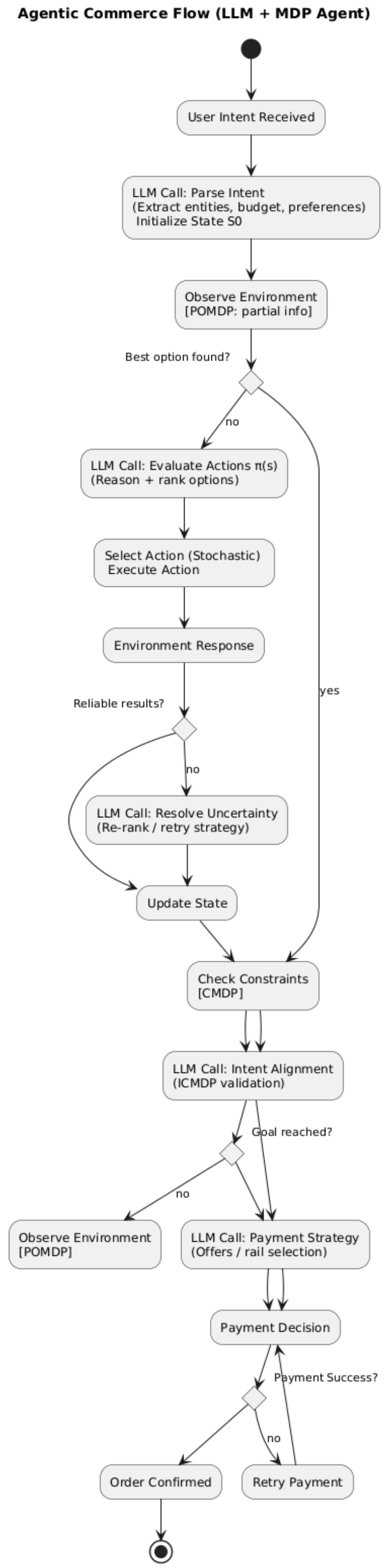

Figure 1: Closed-loop Agentic Commerce decision workflow. The AI agent orchestrates the full purchasing lifecycle—from natural language intent reception to order confirmation—integrating POMDP belief updates (for partial observability), CMDP cost checks (for resource constraints), and ICMDP admissibility validation (for intent fidelity) across distinct functional layers. The diagram instantiates the theoretical ICMDP framework in a concrete, architecturally realizable autonomous system.

Table 2: Comparison of alignment mechanisms.

Property	RLHF	CMDP	ICMDP
Constraint type	Soft (reward shaping)	Expectation	Hard (admissibility)
Intent representation	Implicit ($\hat{R}_\phi$)	Cost thresholds	$\mathcal{I}(s)$ function
Violation guarantee	None (statistical)	In expectation	Zero (per step)
Qualitative intent	Approximate	Limited	Full
Computational cost	Training-time	Policy opt. + Lagrangian tuning	Policy opt. + LLM inference
Verifiability	Low	Moderate	High

7.2 ICMDP vs. RLHF: Structural Distinctions

Table 2 summarizes the key structural distinctions between RLHF, CMDP, and ICMDP.

RLHF and ICMDP address alignment from fundamentally different perspectives: RLHF modifies the reward function [4], whereas ICMDP constrains the action space. The two are complementary: RLHF can refine behavior qualitatively *within* the feasible region defined by ICMDP, yielding a hybrid that separates capability optimization from constraint enforcement [1].

7.3 Failure Modes and Open Problems

LLM misinterpretation. The intent constraint function $\mathcal{I}(s)$ is derived from the probabilistic mapping f_θ. Errors in natural language understanding can produce incorrect constraint sets. Unlike reward errors in RLHF—which degrade performance gradually—constraint errors in ICMDP can cause catastrophic infeasibility, where valid actions are excluded entirely. Formal verification of LLM-extracted constraints against consistent ontologies is a necessary safeguard.

Dynamic intent non-stationarity. Classical MDP formulations assume stationary reward and transition functions. Dynamic intent (Section 4.5) violates this assumption, requiring either full re-optimization or incremental constraint updates. Efficient online algorithms for dynamic ICMDPs are an open research area.

Empty feasible set. If $\mathcal{I}$ is overly restrictive, $\mathcal{A}_{\mathcal{I}}(s) = \emptyset$ for some state s, and (13) is undefined. Resolution requires either constraint relaxation or meta-level user interaction, introducing architectural complexity beyond the base ICMDP.

Computational cost of LLM inference. The proposed framework invokes LLMs for initial intent parsing, intermediate alignment checks, and policy validation. Given the latency and cost characteristics of large-scale models, this creates a bottleneck for real-time systems.

Caching, intent pre-compilation, and approximation strategies can mitigate this, but introduce trade-offs between accuracy, cost, and responsiveness.

7.4 Limitations

The analysis in this paper is purely theoretical. No quantitative policy performance comparison has been conducted; solver scalability has not been empirically tested; and real-world robustness under noisy intent extraction has not been evaluated. Empirical validation in practical Agentic Commerce settings is necessary for a definitive conclusion.

8 Conclusion

We have established that while classical MDP, POMDP, and CMDP frameworks provide progressively richer models for uncertainty and constraint handling, they remain fundamentally limited in their ability to represent explicit, state-dependent human intent. The proposed ICMDP framework addresses this gap by introducing intent as a first-class mathematical constraint that directly governs the admissible action space, shifting the optimization paradigm from reward-driven maximization to intent-preserving decision processes.

Through formal analysis, we have shown that ICMDP and CMDP are expressively incomparable (Proposition 4.1): ICMDP admits policies that no CMDP can enforce via expectation-based constraints, while CMDP captures trajectory-level budget feasibility that no state-local intent filter can replicate. The worked Agentic Commerce example demonstrates architectural realizability and identifies the LLM-based intent parser as a principled, modular component that separates semantic interpretation from stochastic optimization.

Several important directions remain open:

1. Development of efficient solution algorithms for dynamically changing intent constraint sets, including online variants of value iteration and policy gradient methods.

2. Formal verification mechanisms for LLM-derived intent mappings, ensuring consistency and alignment before policy optimization.

3. Hybrid architectures integrating ICMDP with learning-based approaches such as RLHF, where RLHF refines qualitative behavior within the feasible region defined by ICMDP.

4. Empirical validation across large-scale, real-time systems to evaluate computational feasibility, robustness under noisy intent extraction, and alignment fidelity.

Overall, ICMDP offers a theoretically grounded direction for advancing autonomous decision-making systems toward more reliable and human-aligned behavior in increasingly complex and stochastic environments.

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